

A QUASI-MULTINOMIAL TYPE OF CONTINGENCY TABLE

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S U M M A R Y

Exact and asymptotic tests of independence are derived for a contingency table based on a half-multinomial, half-negative-multinomial model. These tests are shown to be identical to the well-known corresponding tests for a multinomial model. Some properties are also considered of the probability distribution on which the present model is based.

1. Introduction

The classical theory of testing independence of variates by means of a contingency table is well-known; the asymptotic χ^2 -test was introduced by Pearson (1900) and Fisher (1922), and the exact test based on the conditional hypergeometric distribution was introduced by Fisher (1935). This theory is based on an assumption that the data are obtained by multinomial sampling.

Steyn (1959) considered a model in which negative multinomial sampling is used and derived χ^2 -tests for independence in the case of his model. These tests, and also an exact test for Steyn's model, considered by Rudolph (1963), differ from the corresponding tests for the multinomial model.

The model to be considered in the present paper is related to the above mentioned two models. It arises by fixing one of the marginal totals instead of the sample size (as in the multinomial case) or one of the cell frequencies (as in the negative multinomial case).

Consider the $h \times k$ -fold table (Table 1) which is obtained by sampling from an unspecified bivariate (infinite) population until m observations belonging to the class Y_k are obtained. Let

p_{ij} ($i=1,2,\dots,h$; $j=1,2,\dots,k$) denote the probability that an observation falls into the category $X_i Y_j$.

TABLE 1

	Y_1	Y_2		Y_k	
X_1	n_{11}	n_{12}		n_{1k}	$n_{1.}$
X_2	n_{21}	n_{22}		n_{2k}	$n_{2.}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_h	n_{h1}	n_{h2}		n_{hk}	$n_{h.}$
	$n_{.1}$	$n_{.2}$		m	$n_{..}$

Since the last observation belongs to the class Y_k the probability of obtaining these cell frequencies is given by

$$(1) \quad \frac{m(n_{..}-1)!}{h \cdot k} \prod_{i=1}^h \prod_{j=1}^{k-1} \frac{n_{ij}!}{p_{ij}^{n_{ij}}}$$

This probability function is simply the product of a multinomial and a negative multinomial probability function which give the joint distributions of the cell frequencies in the k -th column and those in the remaining cells, respectively. That is, (1) may be written as

$$(2) \quad \left[\frac{m!}{h \cdot k} \prod_{i=1}^h \left(\frac{p_{ik}}{p_{..k}} \right)^{n_{ik}} \right] \left[\frac{(m + \sum_{i=1}^h \sum_{j=1}^{k-1} n_{ij} - 1)!}{(m-1)! \prod_{i=1}^h \prod_{j=1}^{k-1} n_{ij}!} p_{..k}^m \prod_{i=1}^h \prod_{j=1}^{k-1} p_{ij}^{n_{ij}} \right]$$

2. Some Properties of the Distribution

Clearly, the probability generating function (p.g.f.) of (1) is

$$(3) \quad (p_{..k} + \sum_{i=1}^{h-1} p_{ik} t_{ik})^m (1 - \sum_{i=1}^h \sum_{j=1}^{k-1} p_{ij} t_{ij})^{-m}$$

Hence the factorial moment generating function (f.m.g.f.) is given by

$$(4) \quad M(\alpha_{11}, \dots, \alpha_{h,k-1}) = (p_{..k} + \sum_{i=1}^{h-1} p_{ik} \alpha_{ik})^m (p_{..k} - \sum_{i=1}^h \sum_{j=1}^{k-1} p_{ij} \alpha_{ij})^{-m}$$

By expansion the factorial moments are obtained and the first and second order moments may, for example, be calculated:

$$(5) \quad \begin{aligned} E(n_{ij}) &= \frac{mp_{ij}}{p_{..k}} \text{ for } i=1, \dots, h; j=1, \dots, k-1 \\ E(n_{ik}) &= \frac{mp_{ik}}{p_{..k}} \text{ for } i=1, 2, \dots, h \text{ with } \sum_{i=1}^h n_{ik} = m \\ \text{cov}(n_{ij}, n_{rs}) &= \begin{cases} \frac{mp_{ij}p_{rs}}{p_{..k}^2} & \text{for } j < k, s < k \\ 0 & \text{for } j < k, s = k \text{ (or } j = k, s < k) \\ -\frac{mp_{ik}p_{rk}}{p_{..k}^2} & \text{for } j = s = k \end{cases} \end{aligned}$$

with $i=1, 2, \dots, h; j=1, 2, \dots, k;$
 $r=1, 2, \dots, h$ and $s=1, 2, \dots, k$

$$\text{var}(n_{ij}) = \begin{cases} \frac{mp_{ij}}{p_{..k}} \left(1 + \frac{p_{ij}}{p_{..k}} \right) & \text{for } j < k \\ \frac{mp_{ik}}{p_{..k}} \left(1 - \frac{p_{ik}}{p_{..k}} \right) & \text{for } j = k \end{cases}$$

The joint distribution of the marginal totals $n_{i.}$ and $n_{.j}$ ($i=1,2,\dots,h$; $j=1,2,\dots,k-1$) is given by

$$(6) \quad \sum_{(x)} \frac{m(m + \sum_{i=1}^h \sum_{j=1}^{k-1} x_{ij} - 1)!}{i! j! x_{ij}!} \prod_{i,j} p_{ij}^{x_{ij}}$$

where here, as in subsequent expressions, $\sum_i$ is written for $\sum_{i=1}^h$; $\sum_j$ is written for $\sum_{j=1}^k$; $\prod_{i,j} = \prod_{i=1}^h \prod_{j=1}^k$ is written for $\prod_{i=1}^h \prod_{j=1}^{k-1}$, and $\sum_{(x)}$ is written for summation over all non-negative integer-valued x_{ij} ($i=1,2,\dots,h$; $j=1,2,\dots,k$) subject to the $(h+k)$ constraints

$$\begin{cases} \sum_i x_{ik} = m \\ \sum_i x_{ij} = n_{.j} & , j=1,2,\dots,k-1 \\ \sum_j x_{ij} = n_{i.} & , i=1,2,\dots,h \end{cases}$$

Expression (6) is the coefficient of $\prod_{i=1}^{h-1} t_i^{n_{i.}} \prod_{j=1}^{k-1} u_j^{n_{.j}}$ in the expansion of the p.g.f.

$$(p_{hk} + \sum_{i=1}^{h-1} p_{ik} t_i)^m (1 - \sum_{j=1}^{k-1} p_{hj} u_j - \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} p_{ij} t_i u_j)^{-m}$$

In the case of a 2×2 table (6) reduces to

$$m(m+n_{.1}-1)! p_{22}^{m-n_{.1}} p_{12}^{n_{.1}} p_{21}^{n_{.1}} \sum_{x=\max\{0, n_{.1}-m\}}^{\min(n_{.1}, n_{.1})} \frac{(p_{11} p_{22} / p_{21} p_{12})^x}{x! (n_{.1}-x)! (n_{.1}-x)! (m-n_{.1}+x)!}$$

which under the hypothesis of independence ($H_0 : p_{ij} = p_{i.} p_{.j}$) simplifies to

$$\begin{aligned} & m(m+n_{.1}-1)! p_{2.}^{m-n_{.1}+n_{.1}} p_{1.}^{n_{.1}} p_{.2}^m p_{.1}^{n_{.1}} \frac{(m+n_{.1})!}{m! n_{.1}! n_{.1}! (m+n_{.1}-n_{.1})!} \\ &= \left[\frac{(m+n_{.1}-1)!}{(m-1)! n_{.1}!} p_{.2}^m p_{.1}^{n_{.1}} \right] \left[\frac{n_{..}!}{n_{1.}! (n_{1.}-n_{1.})!} p_{1.}^{n_{1.}} p_{2.}^{n_{1.}-n_{1.}} \right] \end{aligned}$$

This is the product of a binomial and a negative binomial probability function where the parameter of the binomial distribution $n_{..}$ is not fixed but is a stochastic variable determined by $m+n_{.1}$.

The f.m.g.f. of (6) is given by

$$\begin{aligned} & M(\alpha_1, \dots, \alpha_{h-1}, \beta_1, \dots, \beta_{k-1}) \\ &= (p_{.k} + \sum_{i=1}^{h-1} p_{ik} \alpha_i)^m \{1 - \sum_{j=1}^{k-1} p_{hj} (1+\beta_j) - \sum_{i=1}^{h-1} \sum_{j=1}^{k-1} p_{ij} (1+\alpha_i)(1+\beta_j)\}^{-m} \end{aligned}$$

$$= \left[\sum_{s_{1k}, \dots, s_{h-1,k}} \frac{m! p_{.k}^{(m - \sum_{i=1}^{h-1} s_{ik})} \left(\prod_{i=1}^{h-1} p_{ik} s_{ik} \alpha_i \right)}{(m - \sum_{i=1}^{h-1} s_{ik})! \prod_{i=1}^{h-1} s_{ik}!} \right] \times$$

$$\left[\sum_{\substack{s_{11}, \dots, s_{1,k-1} \\ \dots \dots \dots \\ s_{h1}, \dots, s_{h,k-1}}} \frac{(m + \sum_{i,j=1}^{k-1} s_{ij} - 1)! \left\{ \prod_{i,j=1}^{k-1} p_{ij} s_{ij} \right\} \prod_{i=1}^{h-1} (1 + \alpha_i)^{\sum_{j=1}^{k-1} s_{ij}} \left\{ \prod_{j=1}^{k-1} (1 + \beta_j)^{\sum_{i=1}^{h-1} s_{ij}} \right\}}{(m-1)! \prod_{i,j=1}^{k-1} s_{ij}!} \right]$$

Using the notation $x^{!r}$ to denote $x(x-1)\dots(x-r+1)$ and applying first the binomial and then Van der Monde's expansion it follows that

$$E(n_{.j}^{!r})$$

$$= p_{.k}^m \left[\sum_{\substack{s_{11}, \dots, s_{1,k-1} \\ \dots \dots \dots \\ s_{h1}, \dots, s_{h,k-1}}} \frac{(m + \sum_{i,j=1}^{k-1} s_{ij} - 1)! \left\{ \prod_{i,j=1}^{k-1} p_{ij} s_{ij} \right\} \left\{ \sum_{i,j=1}^{k-1} s_{ij} + r \right\} \prod_{i=1}^{h-1} (r-1) s_{lj} + \dots + s_{1j} \dots s_{hj}}{(m-1)! \prod_{i,j=1}^{k-1} s_{ij}!} \right]$$

$$= p_{.k}^m \{ (m+r-1)^{!r} \frac{p_{ij}^r}{p_{.k}^{m+r}} + (m+r-1)^{!r} \sum_{i \neq j} \frac{p_{ij}^{(r-1)} p_{lj}}{p_{.k}^{m+r}} + \dots + (m+r-1)^{!r} \frac{p_{1j} \dots p_{hj}}{p_{.k}^{m+r}} \}$$

$$= (m+r-1)^{!r} \frac{p_{.j}^r}{p_{.k}^r}$$

$$E(n_{i.}^{!r}) = \sum_{t=0}^r \frac{m^{!t}}{t!} \frac{(m-t)^{!t}}{p_{ik}^t} \left\{ \frac{(m+r-t-1)^{!(r-t)}}{(r-t)!} \left(\prod_{j=1}^{k-1} p_{ij} \right)^{(r-t)} \frac{-(m+r-t)}{p_{.k}^{m+r-t}} \right\} r!$$

$$= p_{.k}^{-r} \sum_{t=0}^r \binom{r}{t} m^{!t} (m+r-t-1)^{!(r-t)} \prod_{i,j=1}^{k-1} p_{ij}^{(r-t)}$$

$$\text{with } x^{!0} = 1 \text{ and } \binom{r}{t} = \frac{r!}{t!(r-t)!}$$

Hence

$$(7) \quad E(n_{.j}) = mp_{.j}/p_{.k} \text{ and } E(n_{i.}) = mp_{i.}/p_{.k}$$

$$(8a) \quad \text{var}(n_{.j}) = mp_{.j}(p_{.j} + p_{.k})/p_{.k}^2 \text{ and } \text{var}(n_{i.}) = mp_{i.}(p_{i.} - 2p_{ik} + p_{.k})/p_{.k}^2$$

with $i=1,2,\dots,h$; $j=1,2,\dots,k-1$.

Similarly,

$$(8b) \quad E(n_{.j} n_{.l}) = \frac{m(m+1)p_{.j}p_{.l}}{p_{.k}^2}; \quad \text{cov}(n_{.j}, n_{.l}) = \frac{mp_{.j}p_{.l}}{p_{.k}^2}$$

with $l=1,2,\dots,k-1$; $l \neq j$.

$$E(n_{i.} n_{r.}) = \{m(m-1)p_{.k}^{m-2} p_{ik} p_{rk}\} p_{.k}^{-m} + \{mp_{.k}^{m-1} p_{ik}\} mp_{.k}^{-m-1} \sum_{j=1}^{k-1} p_{rj} + \{mp_{.k}^{m-1} p_{rk}\} \times \\ mp_{.k}^{-m-1} \sum_{j=1}^{k-1} p_{ij} + p_{.k}^m \{m(m+1)p_{.k}^{-m-2} (\sum_{j=1}^{k-1} p_{ij}) (\sum_{j=1}^{k-1} p_{rj})\} \\ = \frac{m}{2} (mp_{i.} p_{r.} + p_{i.} p_{r.} - p_{ik} p_{rk} - p_{ik} p_{rk})$$

$$(8c) \quad \text{cov}(n_{i.}, n_{r.}) = \frac{m}{2} (p_{i.} p_{r.} - p_{ik} p_{rk} - p_{rk} p_{ik})$$

with $r=1, 2, \dots, h$; $r \neq i$

$$E(n_{i.} n_{.j}) = \{mp_{.k}^{m-1} p_{ik}\} mp_{.k}^{-m-1} \sum_{i=1}^h p_{ij} + p_{.k}^m \{m(m+1)p_{.k}^{-m-2} (\sum_{\ell=1}^{k-1} p_{i\ell}) p_{.j}\} \\ = \frac{m}{2} p_{.j} (mp_{i.} + p_{i.} - p_{ik})$$

$$(8d) \quad \text{cov}(n_{i.}, n_{.j}) = \frac{mp_{.j}}{2} (p_{i.} - p_{ik})$$

In the case of a fourfold table the coefficient of correlation obtained from (8a) and (8d) is

$$= \frac{\sqrt{p_{.1} p_{11}}}{\sqrt{p_{1.} (p_{11} + p_{22})}}$$

Note that zero correlation is not implied by independence of variates. On the other hand, if $p_{11} = 0$ the correlation is zero and the probability function of the marginal totals, given by the coefficient of $t^{n_{1.}} u^{n_{.1}}$ in the expansion of

$(p_{22} + p_{12}t)^m (1 - p_{21}u)^{-m}$, becomes the same as that of the cell frequencies viz. the product of independent binomial and negative binomial distributions with identical parameters m :

$$\phi_o(n_{1.}; m) \psi_o(n_{.1}; m) = \left[\frac{m!}{n_{1.}! (m - n_{1.})!} p^{n_{1.}} q^{m - n_{1.}} \right] \left[\frac{(m + n_{1.} - 1)!}{(m - 1)! n_{1.}!} p^{n_{1.}} q^{m - n_{1.}} \right]$$

where

$$p = p_{12}/p_{.2}, q = 1 - p, p' = p_{21}/p_{1.}, q' = 1 - p' = p_{.2}$$

It may also be verified, in the case of a fourfold table, that the regression equations are

$$(9) \quad E(n_{1.} | n_{.1}) = \frac{mp_{12}}{p_{.2}} + \frac{p_{11}}{p_{.1}} n_{.1}$$

and

$$(10) \quad E(n_{.1} | n_{1.}) = -m + \frac{1}{(1 - p_{21})} \left[\frac{n_{1.} (m + n_{1.} - r)!}{r! 0 (n_{1.} - r)! (m - r)! r!} \left[\frac{p_{12} (1 - p_{21})}{p_{11} p_{22}} \right]^r \right]$$

3. The χ^2 -Test

By property (2), using the χ^2 -expressions for the multinomial and negative multinomial distributions, derived by Pearson (1900) and Steyn (1955) respectively, it follows that

$$(11) \quad \chi^2 = \sum_i \sum_j \frac{\{n_{ij} - m(\frac{p_{i.}}{p_{.k}})\}^2}{\frac{mp_{ij}}{p_{.k}}} - \frac{(m + \sum_{j=1}^{k-1} n_{ij} - \frac{m}{p_{.k}})^2}{m/p_{.k}}$$

has an asymptotic χ^2 -distribution with $hk-1$ degrees of freedom (d.f.).

Under the hypothesis of independence of variates (11) becomes

$$(12) \quad \chi^2 = \sum_i \left[\frac{(n_{ik} - mp_{i.})^2}{mp_{i.}} + \sum_{j=1}^{k-1} \frac{(n_{ij} - mp_{i.}p_{.j}/p_{.k})^2}{mp_{i.}p_{.j}/p_{.k}} \right] - \frac{(m + \sum_{j=1}^{k-1} n_{ij} - m/p_{.k})^2}{m/p_{.k}}$$

From (7) it follows that

$$(13) \quad \left(\frac{\hat{p}_{.j}}{p_{.k}}\right) = \frac{n_{.j}}{m} \quad \text{and} \quad \left(\frac{\hat{p}_{i.}}{p_{.k}}\right) = \frac{n_{i.}}{m}$$

are unbiased estimators of $\frac{p_{.j}}{p_{.k}}$ and $\frac{p_{i.}}{p_{.k}}$, respectively. Hence, using the relationships

$$(14) \quad \begin{cases} p_{i.} = \left(\frac{p_{i.}}{p_{.k}}\right) / \left(\frac{1}{p_{.k}}\right) \\ \frac{1}{p_{.k}} = \sum_i \left(\frac{p_{i.}}{p_{.k}}\right) \end{cases}$$

(12) gives

$$\chi^2 = \left[\sum_i \frac{(n_{ik} - mn_{i.}/n_{..})^2}{mn_{i.}/n_{..}} + \sum_{j=1}^{k-1} \frac{(n_{ij} - n_{i.}n_{.j}/n_{..})^2}{n_{i.}n_{.j}/n_{..}} \right]$$

or, conveniently writing $n_{.k}$ for m ,

$$(15) \quad \chi^2 = \sum_i \sum_j \frac{(n_{ij} - n_{i.}n_{.j}/n_{..})^2}{n_{i.}n_{.j}/n_{..}}$$

Since (13) imposes $(k-1)h$ linear constraints on the random variables n_{ij} and all but one of these constraints are independent, it follows that under the hypothesis of independence (15) has an asymptotic χ^2 -distribution with

$$hk-1 - \{(k-1)h-1\} = (h-1)(k-1) \text{ d.f.}$$

4. The Exact Test

The conditional distribution of the cell frequencies in table 1 for fixed marginal totals $n_{i.}$ and $n_{.j}$ ($i=1, \dots, h$; $j=1, \dots, k-1$) is

obtained from (1) and (6) as

$$(16) \quad p(n_{11}, \dots, n_{hk} | n_{.1}, \dots, n_{.k-1}, n_{1.}, \dots, n_{h.}) \\ = \left[\prod_{i,j} \frac{n_{ij}!}{p_{ij} n_{ij}!} \right] / \left[\sum_{(x)} \prod_{i,j} \frac{x_{ij}!}{x_{ij}!} \right]$$

Under the null hypothesis ($H_0 : p_{ij} = p_{i.} p_{.j}$), using the well-known fact that

$$\sum_{(x)} \left[\prod_{i,j} (x_{ij}!)^{-1} \right] = \frac{n_{..}!}{\left\{ \prod_i n_{i.}! \right\} \left\{ \prod_j n_{.j}! \right\}}$$

where m is denoted by $n_{.k}$ to simplify the notation, (16) reduces to a multivariate hypergeometric probability function

$$(17) \quad \frac{\left\{ \prod_i n_{i.}! \right\} \left\{ \prod_j n_{.j}! \right\}}{n_{..}! \prod_{i,j} n_{ij}!}$$

Thus both the χ^2 -test and the exact test for this model are identical to those of the multinomial model.

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